Case 2: The problem of location that is independent of time

A company tasked with paving a new highway must choose between two locations to set up its construction site.

The cost of transporting material from the construction site to the area to be paved is 20 $/m^3.

The total volume of material to be transported is 50,000 m^3.

The company has committed to completing the project in 4 months (17 weeks, with work taking place 7 days a week).

Information regarding the two alternatives is as follows:

	Construction site A	Construction site B
Average transport distance	3 km	2 km
Monthly rent	10.000 $	20.000 $
Cost of equipment setup and dismantling	150.000 $	200.000 $

- If construction site B is selected, a payment of 35 TL per day will be made for a worker to hold a flag.

ANSWER:

	Construction site A	Construction site B
Rent (total)	4 months x 10.000=40.000$	4 months x 20.000=80.000$
Cost of equipment setup and dismantling	150.000 $	200.000 $
Payment for the flag worker		7 days x 17 weeks x 35 $ =4.165 $
Transportation Cost	50.000m3 x 20$ x 3 km = 3.000.000$	50.000m3 x 20$ x 2 km = 2.000.000$
Total Cost	=3.190.000$	2.334.165$

In this case, construction site B, which has a lower cost, will be chosen.